

IT Research Methodology

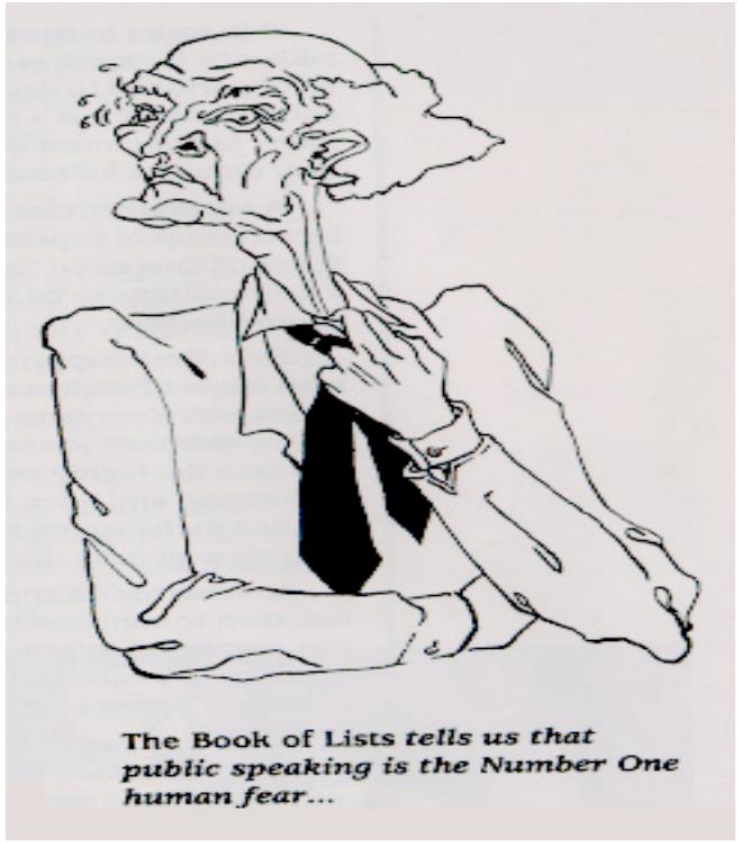
Article Presentation

Image Source: Research Methodology

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Outline

- Before you begin …
- Structuring the Presentation
- Principles of Effective Visual Support
- Delivering the Presentation



❖ Before you begin …

➢ Oral vs. written communication

- ❖ Challenge for the presenter:
  - ✓ Time constraint on information presented
- ❖ Challenge for the audience:
  - ✓ Can’t control rate of presentation to match their comprehension
  - ✓ Can’t re-read sections

Before you begin …

- **Content is the Key!**
  - If there is no content, there is no presentation
- **Know your material and its message**
  - Collect more information than you will use
  - Review the experiment
  - Review graphs and charts
  - Anticipate problem areas

❖ Before you begin …

➢ Ask yourself

- ❖ What is the main point I want to make to my audience?
- ❖ Why is this interesting or important?
- ❖ How do the data support my main point?
- ❖ What part of my story can I tell with the data in the allotted amount of time?

❖ Structuring the Presentation

➢ Tell a story

- ❖ Narrative Structure: beginning, middle and end
- ❖ Show how each section relates to and builds upon the one before it
- ❖ Engage the audience’s interest as they follow the narrative

❖ Structuring the Presentation

➢ Introduction

- ❖ Introduce yourself
- ❖ Give the title (and author) of your talk
- ❖ In one sentence, introduce the central question or problem of the experiment
- ❖ State significance of experiment; why should we care?
- ❖ Briefly explain necessary background
- ❖ Give audience a preview of approach to problem

❖ Structuring the Presentation

➢ Data

- ❖ Forms bulk of presentation
- ❖ Drawn from Methods, Results and Discussion of paper
  - ✓ keep explanation of methods to a minimum only as much as needed to understand results
  - ✓ integrate discussion as you go
- ❖ Data are only worth presenting insofar as they relate to your central question

❖ Structuring the Presentation

➢ Summary

- ❖ What do you want your reader to remember about your talk?
- ❖ Remind your reader of primary findings
- ❖ Explain what these findings contribute to the field

❖ Structuring the Presentation

- **Q & A**
  - ❖ Anticipate questions not covered in the presentation
  - ❖ OK to bring extra slides
  - ❖ OK to acknowledge gaps in expertise
    - ✓ Explain what you do know

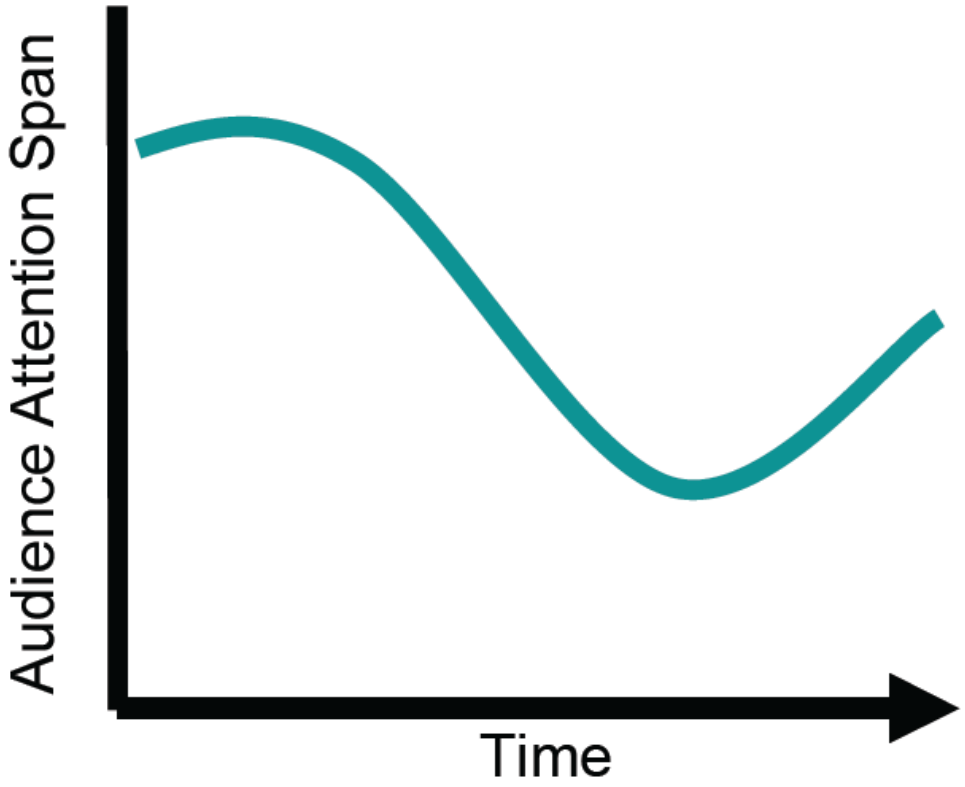
❖ Structuring the Presentation

- **Arrange ideas in a logical sequence**
  - ❖ Most important point first
  - ❖ Emphasize key points as you make them
  - ❖ Provide explicit transitions between points

❖ Structuring the Presentation

➢ Preview and Review

- ❖ Map out goals of the talk in advance
  - ✓ Use topic sentences in body of the talk
- ❖ Summarize
  - ✓ at end of each section
  - ✓ at end of your talk



❖ Principles of Effective Visual Support

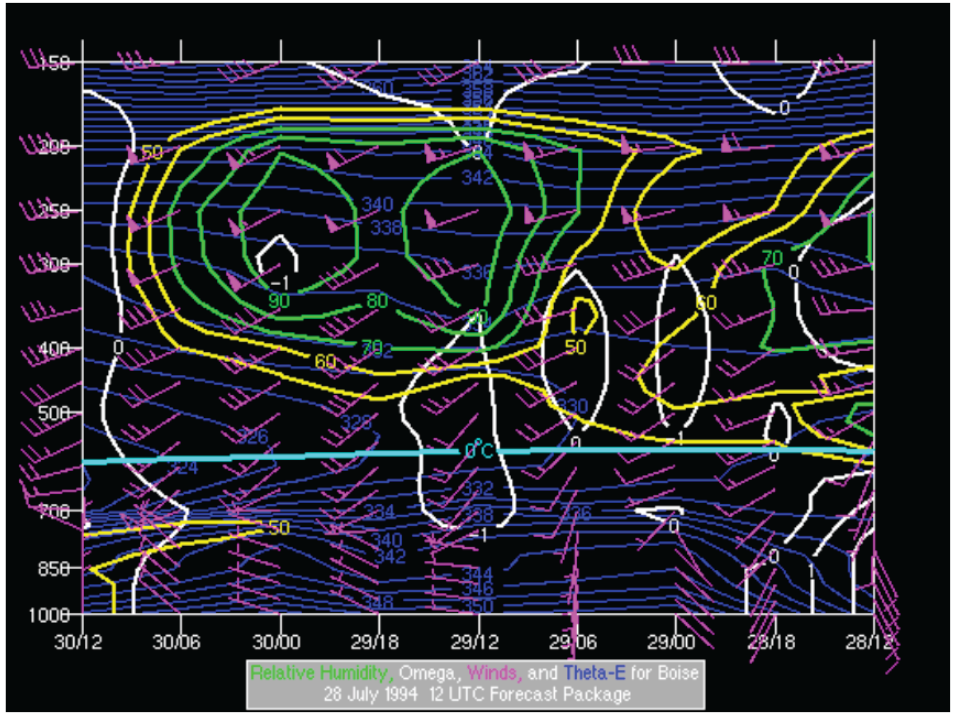
➢ How to Design Effective Slides

- ❖ Limit number of slides
- ❖ Provide explanatory headings for all slides
- ❖ Use illustrations and graphics liberally
- ❖ Minimize text
- ❖ Avoid potentially annoying animation

❖ Principles of Effective Visual Support

➢ Graphics

- ❖ Keep them simple
  - ✓ Average attention span per slide is 8 seconds
- ❖ Title all charts, tables, and diagrams
  - ✓ Clarify the purpose of your slide
- ❖ Use clear, explanatory labels
  - ✓ Make sure to label axes!



❖ Principles of Effective Visual Support

➢ More Design Principles

❖ **Color**

- Be easy on the eyes; don't distract from content
- Avoid low-contrast combinations

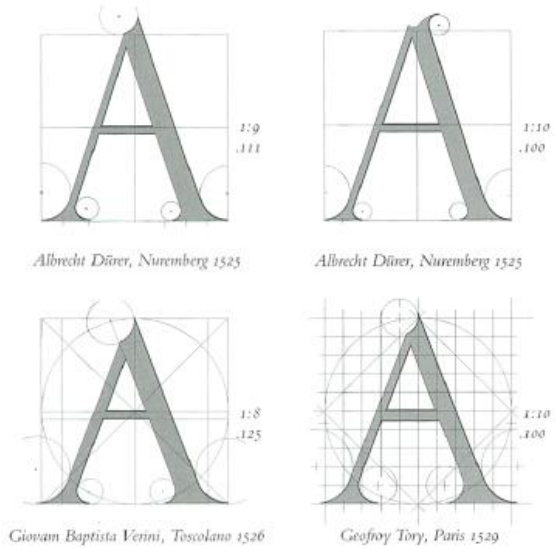
❖ Type

✓ Sans serif headings

✓ Serif bullets (serif "feet" make lines for ease of reading)

✓ Type at least 20-24 pt

✓ Limit upper-case type



❖ Principles of Effective Visual Support

➢ Using graphics in a presentation

What story does this picture tell?

"As shown in Fig. 2, the loss of neuraminidase activity from the supernatant coincides with the disappearance of this 66-kDa protein. This indicates that neuraminidase activity is precipitated via the 66-kDa protein."

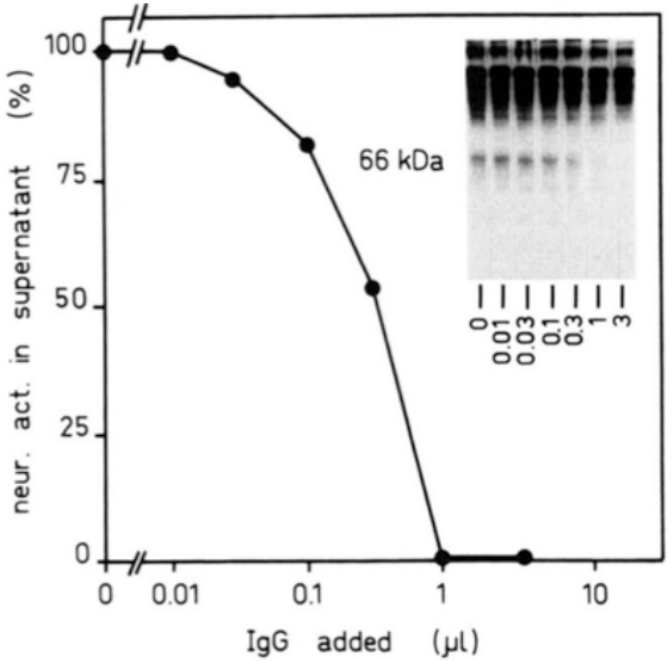
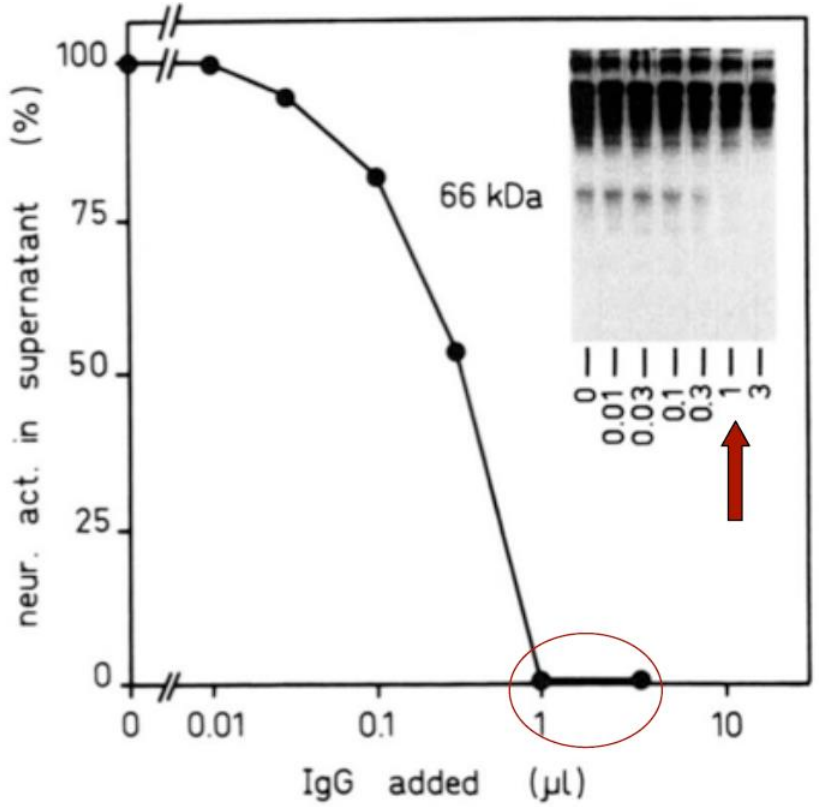


FIG. 2. **Immunotitration of activated and stabilized human placental neuraminidase.** Activated, stabilized neuraminidase was immunoprecipitated from a human glycoprotein preparation with increasing amounts of an IgG preparation prepared from neuraminidase-specific antibodies. Neuraminidase activity was measured in the supernatants. *Inset*, immunoblot analysis of supernatants using neuraminidase-specific antibodies.

From van der Horst GT, Galjart NJ, d'Azzo A, Galjaard H, Verheijen FW. Identification and in vitro reconstitution of lysosomal neuraminidase from human placenta. J Biol Chem. 1989 Jan 15;264(2):1317–1322.

❖ Principles of Effective Visual Support

➢ **Neuraminidase activity is precipitated via 66-kDa**



immunoblot analysis of supernatants

Neuraminidase activity ceases with disappearance of 66-kDa!

❖ Delivering the Presentation

➢ Rehearse!

- ❖ Practice at least 6 times
- ❖ Practice with a colleague for feedback
  - ✓ Is your content clear?
  - ✓ Do you rock, squirm, gesture too much?
  - ✓ Is there room for improvements/adjustments?
- ❖ Time yourself
- ❖ What 3 questions will your audience likely ask?

❖ Delivering the Presentation

➢ On Presentation Day …

- ❖ Arrive early
- ❖ Check equipment and voice projection
- ❖ Bring a backup of your presentation

❖ Delivering the Presentation

➢ How to Connect with the Audience

- ❖ Put yourself in the audience's place
  - ✓ Use everyday language and terms
  - ✓ Explain novel ideas/terms or references
- ❖ Engage the audience
  - ✓ Establish eye contact; look at people
  - ✓ Convey enthusiasm; if you aren't excited about your subject, your audience won't be either
- ❖ A presentation is two-way communication
  - ✓ Pay attention to audience reaction; modify your talk as needed

❖ Delivering the Presentation

➢ Standing

- ❖ Don’t block the screen!
- ❖ Stand at a 45-degree angle to the audience
- ❖ Keep weight evenly dispersed on both feet



❖ Delivering the Presentation

➢ Gesture and Movement

- ❖ Make non-verbal behavior deliberate; avoid extraneous motion
- ❖ Some walking adds variety; too much is distracting
- ❖ Use gestures that complement your speech's content and are natural for you
- ❖ Know what your body language says



❖ Delivering the Presentation

➢ Vocal Issues

- ❖ Volume
  - ✓ Project to back of room: support voice from diaphragm
- ❖ Rate
  - ✓ Speaking at appropriate rate for audience comprehension
  - ✓ Slow down for especially complex or important content
- ❖ Pitch
  - ✓ Keep pitch of your voice at natural level
  - ✓ Avoid “uptalk”



❖ Delivering the Presentation

➢ Handling Anxiety

- ❖ Practice and prepare
- ❖ Focus and center yourself
- ❖ Breathe
- ❖ Have a conversation

❖ Delivering the Presentation

➢ Now What?

- ❖ Get acquainted with the research
- ❖ Design your slides
- ❖ Practice your talk
- ❖ Deliver your talk
- ❖ Meet to review video and slides

Reference

Purpose, Movement, Color: A Strategy for Effective Presentations

- Tom Mucciolo and Rich Mucciolo, MediaNet, Inc., 1994

The Craft of Scientific Presentations

- Michael Alley, Springer, 2005

The Quick and Easy Way to Effective Speaking

- Dale Carnegie, Dale Carnegie Associates, Inc., 1962

The Visual Display of Quantitative Information

- Edward R. Tufte, Graphics Press, 1983

Basic Outline – 10 min presentation

- Background – 3-4 slides
  - e.g. Aims, Hypothesis
- Methods – 4-5 slides
  - e.g. Study Design, Data, Statistical Analysis
- Results – 3-4 slides
- Summary – 1-2 slides
- Limitations/Future Directions – 1-2 slides
- Acknowledgements – 1 slide

Q&A